DINOv3 meets Sketches: A General Sketch-Photo Encoder

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Abstract. Sketches are considered a very relevant means of communication for humans. In fact, these abstract images are our first approach to real objects during our early childhood and allow us to unfold complex processes or transmit complex ideas, facilitating understanding. Therefore, it should also be considered an essential element in computer vision. Beyond sketch-based image retrieval, we can leverage sketches to guide other visual tasks, including segmentation or localization. To this end, it is critical to account for a highly effective sketch-based encoder trained in a bimodal fashion with sketches and photos. However, having sketch-photo pairs is unfeasible, so self or weakly supervised learning is required. In addition, given DINOv3's outstanding performance, it is crucial to leverage it to make it applicable to the sketch-based domain while retaining its learned knowledge. Therefore, in this work, we allow DINOv3 to meet sketches by proposing a new encoder, **skDINO**, that demonstrates general effectiveness across different visual tasks. We evaluate our new encoder in zero-shot sketch-based image retrieval for category (ZS-SBIR) and fine-grain retrieval (ZS-FG-SBIR), achieving new SOTA performance that improves current models by 24.3%, 30.3%, and 15.7% in mAP@200 on Flickr15K, eCommerce, and Sketchy datasets, respectively. For ZS-FG-SBIR, our model reaches an improvement of 15.8% in accuracy in the Sketchy dataset. In addition, we leverage our encoder for sketch-based segmentation, where our model outperforms the state of the art, achieving a new mIoU of 0.7139, an increase of 18.8% over the best known model.

1 Introduction

Sketch-based understanding is a key process for diverse tasks involving human visual perception. Sketches are abstractions of our environment in a minimalist sense. In this work, we consider sketches as simple, handmade drawings of scenes or single objects. Sketches mainly encode shapes of objects in a scene, which have been shown to activate the first layers of our visual cortex in the process of recognizing scenes [20, 40]. Indeed, in the early days of modern computing, the renowned neuroscientists D. Hubel and T. Wiesel [20] showed that the animal visual cortex responds strongly to edge stimuli. Moreover, in more

recent times, Walther et al. [40] used fMRI to demonstrate the semantic power of image contour information in our visual system.

Furthermore, sketching is a natural and primitive means of communication; it is how our ancestors have transmitted ideas, stories, and activities, allowing us to learn from our past. Sketches are also profoundly connected to cognitive development [15]. In fact, infants start to understand their environment through abstract representations (sketches) of real objects [10, 26], and adults use diagrams to unfold and explain complex ideas or processes.

Given the essential role of sketching in visual perception, it is logical to integrate this process into the field of computer vision. In this vein, we have seen an increasing interest in this topic by the vision community, addressing a diversity of sketch-based tasks like sketch classification [44], sketch representation learning [25, 30, 34, 35, 42], sketch-guided object localization [7, 39], sketch-based image retrieval [3, 11, 16, 22, 24, 38, 43], sketch-to-photo translation [1, 12, 28], among others. All those sketch-based understanding tasks have gained interest with time, which can be attributed to the development of modern deep neural networks capable of extracting highly semantic representations from images [6, 32, 33, 37], and the ubiquitous drawing tools that allow us to formulate sketches effortlessly, being even easier than writing text queries.

On the other hand, recent work has made significant advances in foundational models, especially in image encoding. The emergence of high-semantic encoders has improved downstream tasks such as detection [29] and segmentation [21]. Moreover, the use of modern visual encoders has enabled a reduction of the number of trainable parameters by requiring only the adjustment of the decoder head while keeping the encoder frozen for specific tasks. In addition, the growing interest in large multimodal models has enabled the use of multimodal prompting to modulate their behavior without extra training.

Considering the importance of sketch-based interaction, it is essential to develop a general sketch encoder that aligns with modern vision encoders. An effective sketch-based encoder would enable us to tackle a range of vision tasks. For instance, we could use sketches to describe objects we are interested in detecting or segmenting in large image or video catalogs. However, current visual encoders, such as DINOv3 [37], struggle with interpreting sketches (see Figure 4). This problem may be due to the insufficient availability of sketch-photo pairs, which complicates training a bimodal encoder. Consequently, it is crucial to adopt a self-supervised strategy to overcome the absence of sketch-photo pairs to improve a sketch-based encoder.

Therefore, this work bridges the gap between SOTA vision encoders and sketches through our new proposal, skDINO. This encoder builds on DINOv3's knowledge and is adapted to the sketch-based domain, using existing sketch-photo pairs and leveraging images without paired sketches. Our proposal demonstrates high performance across diverse visual tasks, including sketch-based image retrieval and segmentation, beating state-of-the-art performance.

To facilitate reading, we have organized this document into six sections. Section 2 describes the related work. Section 3 presents the proposal describing the

proposed architectures. Section 4 describes the involved experiments. Section 5 discusses the achieved results. Finally, Section 6 provides the final remarks.

2 Related Work

2.1 Self-supervised learning

Modern artificial intelligence models are based on an *encoder-decoder* scheme, where the encoder is responsible for properly representing an image, text, or audio signal into a semantic feature space (a.k.a. latent space). Thus, with a highly semantic encoder, we mostly need to train only the decoder part for downstream tasks, or perhaps a small part of the encoder.

In recent years, we have seen the emergence of increasingly powerful image encoders in computer vision [8, 17, 32, 37]. For an encoder to be truly effective, it is essential to make use of as much available data as possible, even though this data often lacks proper labeling. To this end, self-supervision plays a key role, enabling the model to see diverse images and learn discriminative structures that facilitate downstream inference. Furthermore, the absence of labels can help models generalize better.

For training, a self-supervision approach assumes a pretext task that provides pseudo-labels, allowing the model to learn discriminative patterns. In the context of representation learning, the *de facto* pretext mechanism is **instance discrimination** [8, 17, 32, 37, 45], where each instance is treated as a unique element and the model is trained to discriminate between them.

In terms of training loss, previous models like SimCLR [8] used contrastive learning, but finding effective negative samples remains a major challenge. This problem led to the use of non-contrastive models that require only positive pair samples for training. However, this approach can collapse into a single representation for all images. To avoid this problem, a set of regularization mechanisms is required. Keeping the teacher frozen and using the exponential moving average (EMA) mechanism have been shown to be effective in avoiding the collapsing problem [17, 32].

Other effective regularization strategies for self-supervised representation learning include leveraging patch embeddings, as in iBOT [45]. Here, the visual content of some patches is removed, forcing the model to produce embeddings as if that content had not been removed. More recently, DINOv3 [37] introduced a new loss to mitigate degradation in patch-level consistency. To this end, the model attempts to keep pairwise similarities close to those of early versions during training.

2.2 Encoder Adaptation

To maximize an encoder's discriminative ability in a new context, its parameters should be adjusted. However, we need to be careful about catastrophic forgetting. In fact, it is essential to preserve the visual patterns learned by an encoder

trained on a large amount of data to avoid overfitting and enable generalization. Therefore, instead of adjusting all encoder parameters, adaptation mechanisms focus on training only a small number of them. There are two standard adaptation mechanisms, the first adjusts the parameters of certain layers, a common choice is to adjust the normalization ones [35]. The second approach uses additional neural blocks that modulate the output of the pretrained layers without altering the pretrained encoder [18].

2.3 Self-Supervised Sketch-based Encoders

A sketch-based encoder is the core of sketch-based image retrieval (SBIR), but it extends beyond it. Thus, we are interested in leveraging a sketch-based encoder for diverse tasks, including sketch-based segmentation or detection. In fact, we could regard an input sketch as a prompt for a foundational image model that guides the underlying downstream task. A sketch-based encoder is a bimodal task that involves sketches and regular images (photos), whose representations should be homologated. Therefore, we can also refer to the sketch-based encoder as a sketch-photo encoder, because it should be able to encode a sketch and a photo properly.

For training a sketch-based encoder, and given the bimodal nature of the task, we need access to a large number of sketch-photo pairs, which is unfeasible. Therefore, it is critical to leverage a self-supervision mechanism together with bimodal training [27, 33] to produce a general sketch-based encoder. Thus, "self-supervision" refers to training without paired data, meaning we do not have access to real sketches paired with photos. We only have access to a collection of photos for training (the unpaired dataset). Nevertheless, for evaluation, we should use pairs formed from real sketches, as in real-world practical scenarios.

Unlike supervised sketch-based encoders [11, 35], such as SketchFusion [25], there are only a few works on self-supervised sketch-based image retrieval. For instance, Morales et al. [30] proposed SBIR-BYOL as an extension of BYOL for sketch-image representations. Recently, Campos et al. [4] evaluated different SSL mechanisms for sketch-photo representations, and Saavedra et al. [34] proposed adapting the DINOv2 [32] and CLIP [33] encoders using contrastive loss. However, the proposed SSL models were evaluated only for image retrieval, preventing us from assessing their impact on other vision tasks.

Therefore, inspired by the recent proposals on self-supervised sketch-based representations and modern visual encoders, we propose skDINO, which uses self-supervision to exploit photo catalogs without paired real sketches. Moreover, we also leverage existing sketch-photo pairs [36] to improve abstraction understanding. Thus, this work provides the following contributions:

1. A general sketch-based encoder leveraging DINOv3's knowledge and adjusted by self-supervision and existing sketch-photo pairs.
2. New state-of-the-art performance on zero-shot sketch-based image retrieval, improving current models by 24.3%, 30.3%, and 15.7% on Flickr15K, eCommerce, and Sketchy datasets, respectively

3. New state-of-the-art performance on zero-shot fine-grain sketch-based image retrieval, improving current models by 15.8% in the Sketchy dataset.
4. New state-of-the-art performance on sketch-based segmentation, beating current models with a gain of 18.8%.

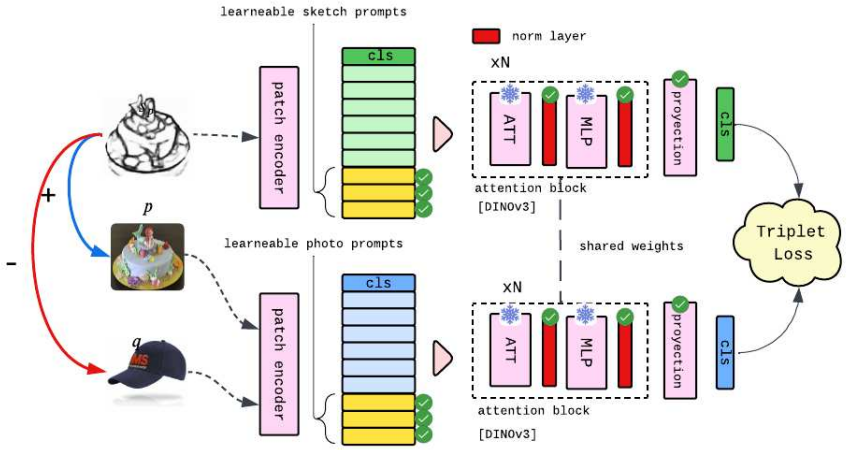


Fig. 1: General scheme of the skDINO encoder, which is based on DINOv3. The blocks marked with a green ball are trainable; the rest keep frozen.

3 skDINO

In this section, we describe in detail the components of skDINO, the general proposed sketch-based encoder. This proposal leverages DINOv3's knowledge and adapts it to understand sketches. Therefore, the main architecture is the same as DINOv3, adding some modifications to facilitate sketch adaptation. Figure 1 depicts the general architecture of skDINO, showing the frozen and learnable layers.

3.1 General architecture

As in DINOv3, the core architecture of skDINO is a ViT-based model. Moreover, given the bimodal nature of our problem, we could use two encoders, as in CLIP. However, instead of two separate encoders —one for sketches and the other for photos — we use a single DINOv3 encoder with most parameters frozen. However, to modulate the encoder's behavior to process a sketch or a photo differently, we use the prompt-based strategy proposed by [35]. Thus, we use two sets

of learnable prompts—one for sketches and the other for photos—concatenated with the initial patch embeddings. At the end of the architecture, we include a trainable projection layer to produce the global representation from the corresponding cls-token.

3.2 Adaptation Mechanisms

We tried with two adaptation strategies. The first follows the proposal of Sain et al. [35], adjusting only the normalization layers of the pretrained encoder. In the second, we apply the LoRA (Low Rank Adaptation) mechanism [18] in the attention blocks, varying the size of the inner projection r.

3.3 Training Loss

We use a contrastive loss following the proposal of Saavedra et al. [34], who showed that the triplet loss based on cosine distance outperforms the InfoNCE approach [31].

3.4 Training without Real Sketches

Having a dataset with a huge amount of sketch-photo pairs is unfeasible. Consequently, it is crucial to define a process to produce pseudo-sketches for each sample image. To this end, we follow the proposal of Morales et al. [30] that leverages semantic contour detection models to simulate a sketch from a photo. In this manner, for each photo p in a batch, we use its contour s_p as the anchor to form the positive pair (s_p, p). The negative pair (s_p, q) is formed by picking randomly another photo q from the batch as shown in Figure 1.

3.5 Training Datasets

We leverage the proposed dataset for sketch-based image retrieval. For training, we use the image catalog of Flickr25K [13] and eCommerce [30]. We selected these datasets because they have low background noise in the photos, enabling us to automatically produce semantic contours. Combining both datasets, we have 75701 photos paired with their corresponding semantic contours, as explained before. Producing semantic contours from noisy images is also an interesting topic that could yield benefits for a better sketch-based encoder, but it remains a challenging problem [1], thus it is beyond the scope of this work.

However, it is also important to allow the model to learn real sketch abstraction. To this end, we leverage a portion of the Sketchy dataset [36] during training. In particular, we use 49176 sketch-photo pairs distributed into 100 categories.

Considering that we have semantic contours and real sketches, we apply a curriculum learning strategy [2]. First, we train our model using the semantic contours from the Flickr25K and eCommerce datasets for 50 epochs. After that, we train another 50 epochs with the Sketchy dataset.

3.6 Parameter Settings

We trained our models with the AdamW optimizer and a batch size of 64. We use two different learning rates: $lr_a = 1e-4$ for the normalization layers during DinoV3 adaptation, and $lr = 1e-4$ for the proposed learnable prompts and projection layer. In addition, for training and inference, photos and sketches are resized to 224×224 pixels.

4 Experiments

Most sketch-based encoders are evaluated in sketch-based image retrieval, enabling us to assess the effectiveness of the shared latent space learned across sketches and photos. In this work, beyond the global representation, we are also interested in assessing the coherence and effectiveness of the learned patch embeddings of the proposed encoder. To this end, we bring skDINO to the image segmentation problem, where sketches define target objects, as in the work of Subhadeep et al. [25].

4.1 Sketch-based Image Retrieval

Given a sketch as a query, a sketch-based image retrieval (SBIR) method aims to retrieve images or photos that resemble the input query. In this context, the key component is the sketch-based encoder, which should represent both the query sketch and the catalog photos in a shared space. With an effective encoder, retrieval is performed via a similarity search using a distance function.

In this vein, we are interested in evaluating our proposed models under the ZS-SBIR protocol, which means the evaluation does not use samples from categories seen during training. To this end, we use two datasets: Flickr15K [19], with 33 unseen classes and 329 sketches, and eCommerce [4], with 665 sketches, as used previously by self-supervised approaches in this context [4, 34]. In addition, we are also interested in comparing our proposal with the SketchFusion method [25], an SOTA supervised sketch-based encoder. To this end, we include the Sketchy [36] dataset for evaluation, with 25 unseen classes and 3002 sketches, which also enables us to evaluate ZS-FG-SBIR. We will present quantitative results using the mean average precision (mAP) metric and a recall-precision recall for ZS-SBIR and accuracy for ZS-FG-SBIR. Furthermore, we present variations of skDINO as described below:

- **skDINO-A**: This follows the same training protocol described by Saavedra et al. [34], where a sketch-based encoder is trained in Flickr25K and eCommerce, separately, to be evaluated in Flickr15K and eCommerce-test, respectively. We incorporate this version to be comparable with other SOTA methods [4, 34].
- **skDINO-B**: This version merges the training dataset from version A into a single dataset, enabling a single skDINO to be evaluated across diverse testing datasets. Given the good results achieved by this version, the remaining versions will use the merged training dataset proposed here.

- **skDINO-C**: We observed that reducing the learning rate for the normalization layers and increasing it for the learnable prompts improves performance. Therefore, in this version, we use $lr_a = 5e-6$ and $lr = 5e-2$. Regarding the training dataset, we use the same protocol as in version B.
- **skDINO-D**: This version adapts the encoder's attention blocks using LoRA [18], applying a low-rank update with rank $r = 16$. As in skDINO-B, it is trained on the combined dataset, and it follows the same learning-rate configuration used in skDINO-C.
- **skDINO-E**: This version follows the same setting as skDINO-D but increases the LoRA rank to $r = 64$, thereby exploring a higher adaptation capacity while keeping the learning-rate configuration unchanged.
- **skDINO-F**: This version follows the same setting as skDINO-C but incorporates sketch-photo pairs from the Sketchy dataset. For training, we use the curriculum learning described in Section 3.5.

4.2 Sketch-based Segmentation

We have observed, from the cross-attention map, that the shared space inferred by skDINO can also be exploited for object localization. For instance, Figure 2 shows two examples of the correlation between the sketch's cls-token with the photo's patch embeddings. The maximum correlation (in red) occurs in the region where an object similar to the input query appears. This fact inspired us to move skDINO to the sketch-based image segmentation task, the finest localization task.

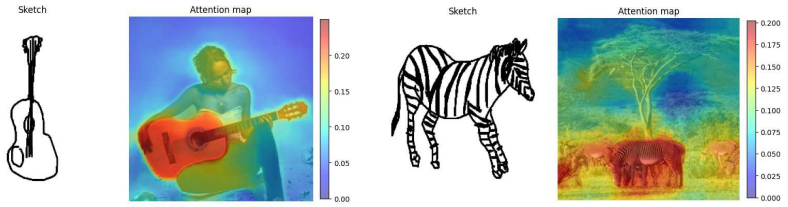


Fig. 2: Correlation between the sketch cls-token against the photo patch embeddings. High correlation is marked with red color. We note the of skDINO for object localization guided by sketches.

Therefore, we evaluate the effectiveness of the proposed skDINO encoder for image segmentation guided by sketches. To this end, we keep the skDINO encoder completely frozen and incorporate a small, mask-based segmentation decoder inspired by modern segmentation models [9,21]. The decoder parameters are the only ones adjusted during training. Figure 3 illustrates the proposed segmentation architecture, which works in a bimodal fashion. The sketch guides the segmentation process, ensuring that the segmented regions correspond to the input sketch.

According to Figure 3, we first obtain the patch embeddings and the cls-token for the query sketch and the target photo. These embeddings are then combined through a bidirectional cross-attention module. The tensor of image embeddings is spatially upsampled using the AnyUp model [41]. On the other hand, we use the sketch cls-token after applying cross-attention. At the end, we use the cls-token as a mask-embedding, correlated with the image embedding, to produce the segmentation mask. In addition, we use the same cls-token to predict a score of the occurrence of the sketch in the input image.

To train and evaluate the segmentation model, we use the dataset proposed by Subhadeep et al. [25]. This dataset consists of 1040 sketch-photo pairs, distributed across 104 classes, with 10 pairs per class.

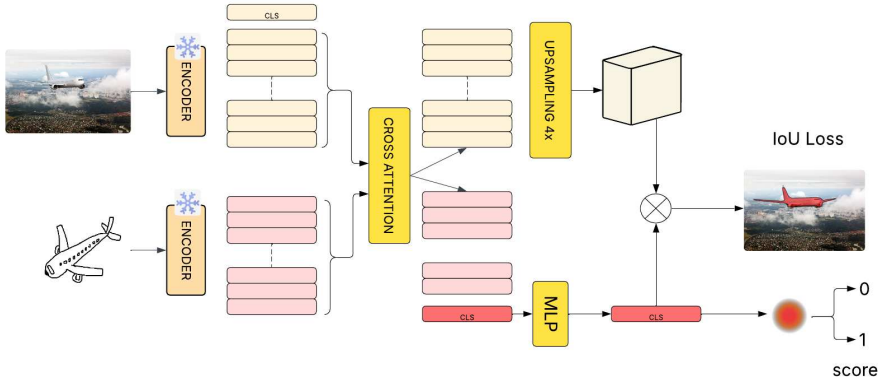


Fig. 3: Sketch-based Segmentation Architecture. Here, skDINO (the encoder) remains frozen, training only the mask-based segmentation decoder.

5 Results and Discussion

5.1 Sketch-based Image Retrieval

ZS-SBIR: We first discuss the performance of skDINO for zero-shot image retrieval using sketches as queries (ZS-SBIR). Table 1 shows the performance of different self-supervised sketch-based encoders for this task. We present results in Flickr15K and eCommerce, following the evaluation protocol proposed by Sain et al. [35] and then used by Saavedra et al. [34] to evaluate self-supervised sketch encoders (S3BIR). Here, we show mAP@all (overall map) and mAP@200 (where ranking is cut off at position 200).

To the best of our knowledge, the best sketch-based encoder is S3BIR-DINOv2 [34], which uses DINOv2 [32] as the encoder. This method achieves an overall mAP of 0.6110 for Flickr15K and 0.4437 for eCommerce. On the other

hand, all our proposals, skDINO-X, beat the current state of the art in terms of mAP@200. In this vein, skDINO-C achieves the best performance among models trained only on semantic contours, with overall mAPs of 0.7356 on Flickr15K and 0.6317 on eCommerce. However, skDINO-F, which additionally uses sketch-photo pairs, is the winner among our proposals, achieving an overall mAP of 0.7775 on Flickr15K and 0.5633 on eCommerce. These new results represent increases in performance of 27.3% for the first evaluated dataset and 27.0% for the second. It is also important to note that pretrained DINOv3 performs worse, suggesting that our method effectively brings DINOv3 closer to the sketch-based domain.

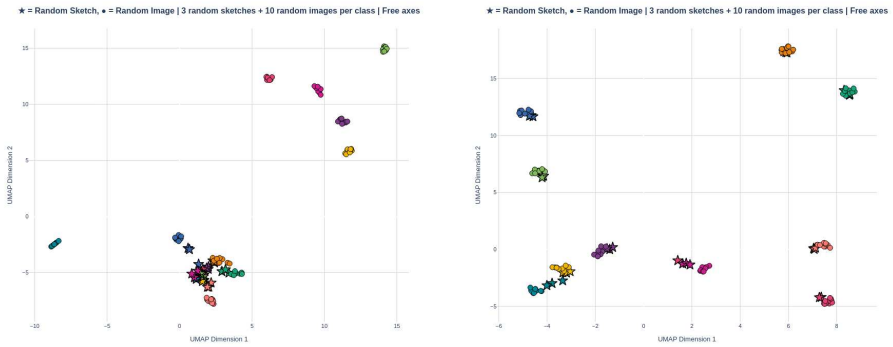
Therefore, using a combined training dataset is relevant to expose the model to a broad spectrum of images, which favors generalization. In addition, the experiments show that the most critical component in adaptation is the learnable prompts, which, when trained with a higher learning rate, improve performance. On the other hand, reducing the learning rate for the layer normalization does not affect the performance.

2*Encoders	Flickr15K		eCommerce	
	mAP@all	mAP@200	mAP@all	mAP@200
CLIP (pretrained)	0.2164	0.2877	0.1221	0.1499
S3BIR-CLIP [34]	0.5403	0.6267	0.4402	0.5290
DINOv2 (pretrained)	0.2106	0.1809	0.1502	0.1804
S3BIR-DINOv2 [34]	0.6110	0.6550	0.4437	0.4911
DINOv3 (pretrained)	0.5087	0.5544	0.4700	0.5452
skDINO-A (ours)	0.6402	0.6940	0.5271	0.6005
skDINO-B (ours)	0.7107	0.7663	0.5324	0.6238
skDINO-C (ours)	0.7356	0.8005	0.5301	0.6317
skDINO-D (ours)	0.7234	0.7779	0.5186	0.6227
skDINO-E (ours)	0.6032	0.7326	0.4225	0.5592
skDINO-F (ours)	**0.7775**	**0.8144**	**0.5633**	**0.6398**

Table 1: Performance comparison between our proposed encoders for ZS-SBIR and the pretrained versions of CLIP, DINOv3, S3BIR-CLIP, and S3BIR-DINOv2. Our proposals A, B, C, and D beat the state of the art, with skDINO-F performing better, achieving new mAP@all of 0.7775 on Flickr15K and 0.5633 on the eCommerce dataset.

Moreover, Figures 4 (A) and 4 (B) show the embedding space inferred by DINOv3 and skDINO-C, respectively. For each plot, we have different image classes differentiated by color. We also use different symbols to represent a sketch or a photo. A star represents sketches, and a circle represents photos. Figure 4 (A) shows the difficulty of DINOv3 to represent sketches. All the sketches form an independent cluster, far from the rest of the images. Our new model keeps related images grouped while placing sketches close to their corresponding groups.

ZS-FG-SBIR: We also present results for zero-shot fine-grain SBIR (ZS-FG-SBIR) using the Sketchy dataset. Table 2 presents the results for both ZS-SBIR



(A) (B)

Fig. 4: (A) UMAP projection of DINOv3 embeddings for sketches and photos. (B) UMAP projection of **skDINO-C** embeddings for sketches and photos. The different colors indicate different categories. In addition, circle-like shapes represent photos and star-like shapes represent sketches.

and ZS-FG-SBIR. In the first case, skDINO-F beats SketchFusion [25], gaining 15.7% in mAP@200. For the fine-grain task, our model achieves an improvement of 15.8% in Acc@1.

2*Encoders	ZS-SBIR		ZS-FG-SBIR	
	mAP@all	mAP@200	Acc.@1	Acc.@5
SketchFusion [25]	-	0.7610	0.3301	0.6792
DINOv3 (pretrained)	0.6965	0.7061	0.1895	0.4667
skDINO-F (ours)	**0.8789**	**0.8803**	**0.3821**	**0.7199**

Table 2: Results on Sketchy dataset [36] for ZS-SBIR and ZS-FG-SBIR tasks. For ZS-FG-SBIR we use accuracy as in SketchFusion [25]. Our results demonstrate the superiority of skDINO over the other proposals.

In addition, Figures 5, and 6 illustrate the qualitative performance of our proposal compared to DINOv3 on the Flickr15K, eCommerce, and Sketchy datasets. For each query, we present the most similar photos, arranged from top to bottom.

To complement our results, Table 3 presents ZS-FB-SBIR performance on the ShoeV2 and ChairV2 datasets. Here, we note that our method beats the state of the art on ChairV2, even though ours has not used either of the two datasets for training. For ShoesV2, we are competitive with Sketch2Word [24], but far from AbstractAway [23]. However, this comparison is unfair, as Sketch2Word and AbstractAway are not zero-shot models; they were trained with ShoeV2 and ChairV2 datasets, respectively. Furthermore, a low performance on specific datasets remains common for foundational models, whose performance strongly

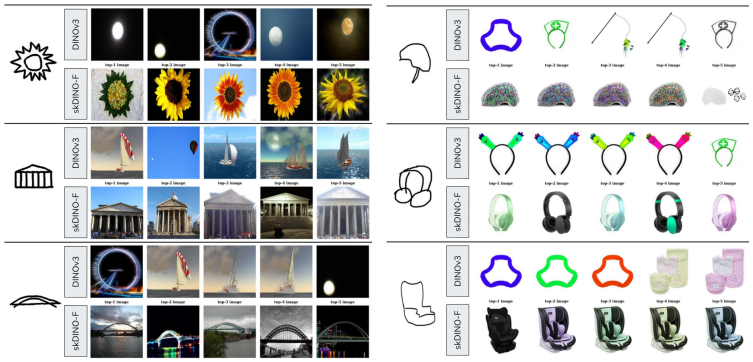


Fig. 5: Examples of ZS-SBIR over Flick15K (left) and eCommerce (right) dataset using DINOv3 (first row) and skDINO (second row).

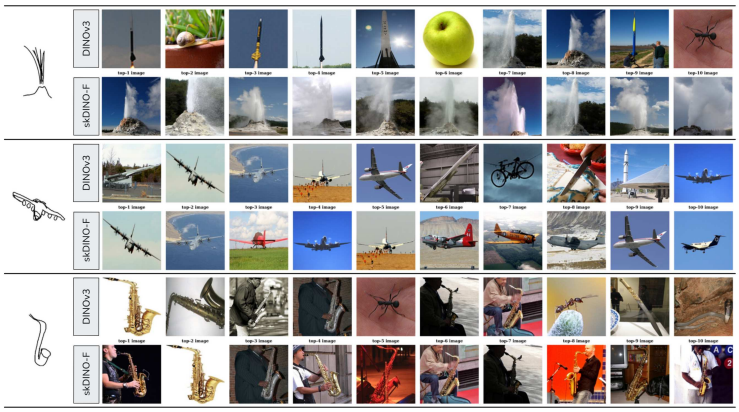


Fig. 6: Examples of ZS-FG-SBIR over Sketchy dataset using DINOv3 (first row) and skDINO (second row).

depends on dataset variability. Our proposal is not unaware of this problem. However, the use of pseudo-sketches can alleviate the lack of real sketches, enabling us to leverage diverse image repositories (without sketches) for training.

5.2 Sketch-based Segmentation

To evaluate this task, we follow the protocol presented in SketchFusion [25], reporting mean IoU (mIoU) and per-pixel accuracy (pAcc). In addition, we also compare our results with those reported by Subhadeep et al. [25].

Thus, our results significantly outperform prior work, achieving new mIoU and mACC of 0.7139 and 0.9396, respectively. Our latest performance results in an increase of 18.8% in mIoU and 25.4% in pAcc. These results demonstrate the effectiveness of the proposed encoder (skDINO) for tasks beyond sketch-based image retrieval.

2*Encoders	ShoeV2		ChairV2	
	Acc@1	Acc@5	Acc@1	Acc@5
Sketch2Word [24]	-	0.4730	-	0.7350
AbstractAway [23]	**0.4530**	**0.7730**	0.7210	0.8090
DINOv3 (pretrained)	0.1712	0.4129	0.5884	0.8360
skDINO-F (ours)	0.1727	0.4234	**0.7267**	**0.9389**

Table 3: FG-SBIR performance on ShoeV2 and ChairV2 datasets. Our method achieves the best result on ChairV2 without using such a dataset for training. However, our method struggles on ShoeV2, where the best model relies on supervision from that dataset.

Methods	**mIoU**	**pAcc.**
SketchFusion [25]	0.6012	0.7491
Proposed (skDINO)	**0.7139**	**0.9396**

Table 4: Results for *sketch-based image segmentation*. We compare our proposal against SketchFusion [25], showing a significant improvement.

Moreover, to visualize the semantic content of the embeddings produced by skDINO, Figure 7 shows the attention-based visualization. Here, we use the cls-token embedding of the input sketch that is correlated with the patch embeddings computed from the target photos. We observe a strong correlation (in red) when a similar object to the sketch appears in the image. When there is no similar object to the query in the photo, the correlation is almost null.

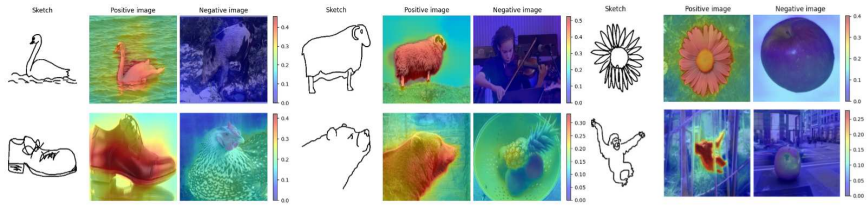


Fig. 7: Cosine similarity maps between sketch cls-tokens and image patch tokens from the pretrained skDINO encoder.

Finally, Figure 8 illustrates qualitative results on the skech-based segmentation achieved by our proposal.

5.3 Patch Embedding Correspondences

To complement our previous results, we also qualitatively demonstrate the capacity of our approach to find local correspondences between a sketch and a photo.

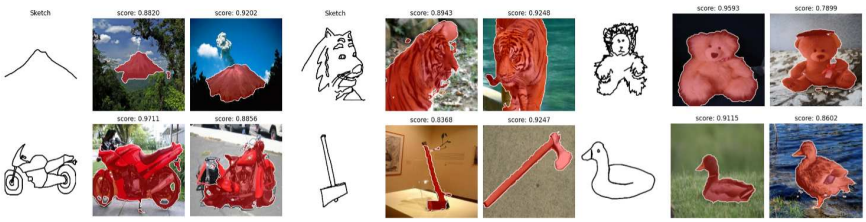


Fig. 8: Qualitative segmentation results of skDINO guided by sketch queries. The score indicates the model's confidence in the presence of the queried object within the image.

Figure 9 illustrates the correspondences between patch embeddings computed by skDINO using a sketch-photo pair. In this case, finding the proper correspondences is achieved using the Hungarian method, commonly applied for image detection [5] or segmentation [9]. In addition, to keep coherent correspondences, we use RANSAC [14].

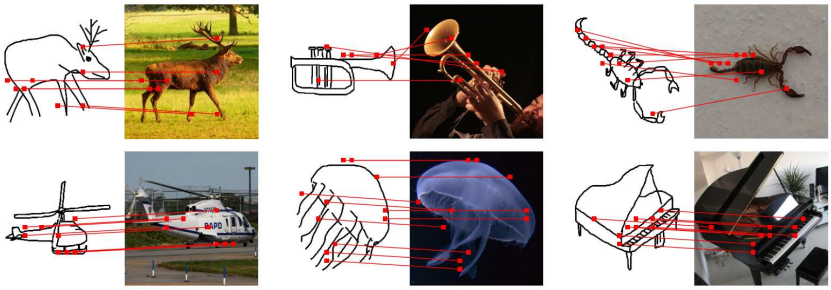


Fig. 9: Examples of sketch-photo correspondences achieved by our proposal skDINO-C. The correspondence is determined by the Hungarian method using the similarity between the patch embeddings from the two involved images.

6 Conclusions

This work introduced skDINO, a new state-of-the-art sketch-based encoder. This encoder leverages DINOv3's knowledge and uses self-supervision to leverage photos without paired sketches. In addition, our proposal leverages sketch-photo pairs from the Sketchy dataset.

We demonstrate the effectiveness of our encoder, skDINO, across different sketch-based tasks, including zero-shot SBIR, category (ZS-SBIR) and fine-grain (ZS-FG-SBIR) retrieval, and sketch-based segmentation. For ZS-SBIR, our model beats prior work, increasing mAP@200 by 24.3%, 30.3%, and 15.7% on

the Flickr15K, eCommerce, and Sketchy datasets, respectively. For the fine-grain task, our proposal achieves a 15.8% gain in accuracy in the Setkchy dataset. Our model also beats the state of the art in sketch-based image segmentation, achieving a new mIoU of 0.7139, an 18.8% increase over prior work.

Finally, we qualitatively demonstrate skDINO's ability to establish appropriate correspondences between sketches and photos. This property could enable skDINO to be leveraged for more challenging problems in the future, such as object search in the wild guided by sketches, considering finer attributes like object pose and orientation.

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